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DSA0105-Object Oriented Programming with C++

CO2-AT1 Scenario Based Question

Question-1:

Define a class named movieMagic with the following description: (20)

Instance variables/data members:

int year – to store the year of release of a movie

String title – to store the title of the movie.

float rating – to store the popularity rating of the movie.

(minimum rating = 0.0 and maximum rating = 5.0)

Member Methods:

(i) void accept() To input and store year, title and rating.

(ii) void display() To display the title of a movie and a message based on the rating as per the table below.

RATING	MESSAGE TO BE DISPLAYED
0.0 to 2.0	Flop
2.1 to 3.4	Semi-hit
3.5 to 4.5	Hit
4.6 to 5.0	Super Hit

Write a main method to create an object of the class and call the above member methods.

PROGRAM

```
#include <iostream>

using namespace std;

class movieMagic
{
    int year = 2025;
    string title = "Avatar";
    float rating = 4.8;

public:
    void accept()
    {
        // Values already assigned
    }

    void display()
    {
        cout << "Movie Title: " << title << endl;

        if (rating <= 2.0)
            cout << "Flop";
        else if (rating <= 3.4)
            cout << "Semi-hit";
        else if (rating <= 4.5)
            cout << "Hit";
        else
            cout << "Super Hit";
    }
}
```

```

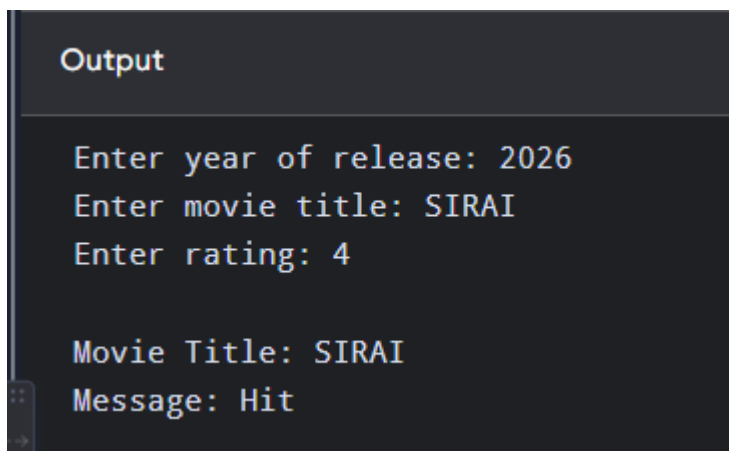
    }
};

int main()
{
    movieMagic obj;
    obj.accept();
    obj.display();

    return 0;
}

```

OUTPUT:



```

Output

Enter year of release: 2026
Enter movie title: SIRAI
Enter rating: 4

Movie Title: SIRAI
Message: Hit

```

Question-2:

Define a class named BookFair with the following description: (20)

Instance variables/Data members:

String Bname – stores the name of the book.

double price – stores the price of the book.

Member Methods:

(i) void Input() – To input and store the name and the price of the book.

(ii) void calculate() – To calculate the price after discount. Discount is calculated based on the following criteria.

PRICE

Less than or equal to Rs 1000

More than Rs 1000 and less than or equal to Rs 3000

More than Rs 3000

DISCOUNT

2% of price

10% of price

15% of price

(iv) void display() – To display the name and price of the book after discount.

Write a main method to create an object of the class and call the above member methods.

PROGRAM

```
#include <iostream>
```

```
using namespace std;
```

```
class BookFair
```

```
{
```

```
    string Bname = "C++ Programming";
```

```
    double price = 2500;
```

```
public:
```

```
    void Input()
```

```
    {
```

```
        // Values already assigned
```

```
    }
```

```
    void calculate()
```

```
    {
```

```
        if (price <= 1000)
```

```
            price = price - (price * 2 / 100);
```

```
        else if (price <= 3000)
```

```
            price = price - (price * 10 / 100);
```

```
        else

            price = price - (price * 15 / 100);

    }

    void display()

    {

        cout << "Book Name: " << Bname << endl;

        cout << "Price after discount: Rs. " << price;

    }

};

int main()

{

    BookFair obj;

    obj.Input();

    obj.calculate();

    obj.display();

    return 0;

}
```

OUTPUT:

```
Book Name: C++ Programming
Price after discount: Rs. 2250
```

Question-3:

Define a class Electric Bill with the following specifications:

Instance Variable/ data member:

String n – to store the name of the customer

int units – to store the number of units consumed

double bill – to store the amount to paid

Member methods:

Void accept() – to accept the name of the customer and number of units consumed

Void calculate() – to calculate the bill as per the following tariff :

Number of units Rate per unit

First 100 units Rs.2.00

Next 200 units Rs.3.00

Above 300 units Rs.5.00

A surcharge of 2.5% charged if the number of units consumed is above 300 units.

Void print() – To print the details as follows :

Name of the customer

Number of units consumed

Bill amount

Write a main method to create an object of the class and call the above member methods

PROGRAM

```
#include <iostream>
```

```
using namespace std;
```

```
class ElectricBill
```

```
{
```

```
    string n = "Deepthi";
```

```
int units = 350;
```

```
double bill;
```

```
public:
```

```
void accept()
```

```
{
```

```
    // Values already assigned
```

```
}
```

```
void calculate()
```

```
{
```

```
    if (units <= 100)
```

```
        bill = units * 2.00;
```

```
    else if (units <= 300)
```

```
        bill = (100 * 2.00) + ((units - 100) * 3.00);
```

```
    else
```

```
{
```

```
        bill = (100 * 2.00) + (200 * 3.00)
```

```
            + ((units - 300) * 5.00);
```

```
        bill = bill + (bill * 2.5 / 100);
```

```
    }
```

```
}
```

```
void print()
```

```
{
```

```
    cout << "Name of the customer: " << n << endl;
```

```
        cout << "Number of units consumed: " << units << endl;

        cout << "Bill amount: Rs. " << bill << endl;
    }
};

int main()
{
    ElectricBill obj;

    obj.accept();
    obj.calculate();
    obj.print();

    return 0;
}
```

OUTPUT:

```
Name of the customer: Deepthi
Number of units consumed: 350
Bill amount: Rs. 1076.25
```

Question-4:

Design a class RailwayTicket with the following description: (20)

Instance variables/data members:

String name: to store the name of the customer.

String coach: to store the type of coach customer wants to travel. long mobno: to store customer's mobile number.

int amt: to store basic amount of ticket.

int totalamt: to store the amount to be paid after updating the original amount.

Methods:

void accept(): to take input for name, coach, mobile number and amount.

void update(): to update the amount as per the coach selected. Extra amount to be added in the amount as follows:

Type of coaches	Amount
First_AC	700
Second_AC	500
Third_AC	250
sleeper	None

void display(): To display all details of a customer such as name, coach, total amount and mobile number.

Write a main() method to create an object of the class and call the above methods.

PROGRAM

```
#include <iostream>
```

```
using namespace std;
```

```
class RailwayTicket
```

```
{
```

```
    string name = "Deepthi";
```

```
    string coach = "First_AC";
```

```
    long mobno = 9876543210;
```

```
    int amt = 1000;
```

```
    int totalamt;
```

public:

void accept()

{

 // Values already assigned

}

void update()

{

 if (coach == "First_AC")

 totalamt = amt + 700;

 else if (coach == "Second_AC")

 totalamt = amt + 500;

 else if (coach == "Third_AC")

 totalamt = amt + 250;

 else

 totalamt = amt;

}

void display()

{

 cout << "Name: " << name << endl;

 cout << "Coach: " << coach << endl;

 cout << "Mobile Number: " << mobno << endl;

 cout << "Total Amount: Rs. " << totalamt << endl;

}

};

int main()

{

```
RailwayTicket obj;
```

```
obj.accept();
```

```
obj.update();
```

```
obj.display();
```

```
return 0;
```

```
}
```

OUTPUT:

```
Name: SRI  
Coach: First_AC  
Mobile Number: 9876543210  
Total Amount: Rs. 1700
```

Question-5:

Design a class named ShowRoom with the following description: (20)

Instance variables/data members:

String name: to store the name of the customer.

long mobno: to store the mobile number of the customer.

double cost: to store the cost of the items purchased.

double dis: to store the discount amount.

double amount: to store the amount to be paid after discount.

Member methods:

void input(): to input customer name, mobile number, cost.

void calculate(): to calculate discount on the cost of purchased items, based on the following criteria:

COST DISCOUNT (IN PERCENTAGE)

Less than or equal to Rs. 10000 5%

More than Rs. 10000 and less than or equal to Rs. 20000 10%

More than Rs. 20000 and less than or equal to Rs. 35000 15%

More than Rs. 35000 20%

void display(): to display customer name, mobile number, amount to be paid after discount.

Write a main() method to create an object of the class and call the above member methods.

PROGRAM

```
#include <iostream>
```

```
using namespace std;
```

```
class ShowRoom
```

```
{
```

```
    string name = "Deepthi";
```

```
    long mobno = 9876543210;
```

```
    double cost = 25000;
```

```
    double dis, amount;
```

```
public:
```

```
    void input()
```

```
    {
```

```
        // Values already assigned
```

```
    }
```

```
    void calculate()
```

```
    {
```

```
        if (cost <= 10000)
```

```
        dis = cost * 5 / 100;
    else if (cost <= 20000)
        dis = cost * 10 / 100;
    else if (cost <= 35000)
        dis = cost * 15 / 100;
    else
        dis = cost * 20 / 100;

    amount = cost - dis;
}

void display()
{
    cout << "Customer Name: " << name << endl;
    cout << "Mobile Number: " << mobno << endl;
    cout << "Amount to be paid: Rs. " << amount << endl;
}

};

int main()
{
    ShowRoom obj;

    obj.input();
    obj.calculate();
    obj.display();

    return 0;
}
```

OUTPUT:

```
Customer Name: SRI  
Mobile Number: 9876543210  
Amount to be paid: Rs. 21250
```

